

Explainable AI in Frequency-Domain CNNs: Theory, Attribution, and Alternative Explanation Methods

Contents

1	Introduction	2
2	Key Principle: Attribution Must Follow the Inverse Transform	2
2.1	Problem Setup	2
2.2	Gradient Propagation and the Adjoint Operator	2
2.3	Interpretation	3
3	Gradient-Based Explainability in Frequency Domain	3
3.1	Pipeline	3
3.2	Why This Works	3
3.3	Important Properties	3
4	Alternative Explainability Methods for Frequency-Based Models	4
4.1	1. Frequency Perturbation / Occlusion	4
4.2	2. Energy-Based Attribution	4
4.3	3. Spectral TCAV (Concept-Based)	5
4.4	4. Frequency Response Analysis	5
4.5	5. Wavelet-Based Explainability	5
4.6	6. Information-Theoretic Attribution	6
4.7	7. Hybrid Spatial–Frequency XAI	6
5	Comparison of Methods	6
6	Best-Practice Recommendation	6
7	Key Insight	7

1 Introduction

In frequency-domain deep learning, convolutional neural networks (CNNs) operate on transformed representations of images, such as Fourier or cosine coefficients. While this representation offers advantages in robustness and expressivity, it complicates explainability. Unlike spatial CNNs where explanations directly correspond to pixel regions, frequency-based models encode information globally and implicitly.

This document presents a complete theoretical and practical analysis of explainability in frequency-domain CNNs, including:

- Why attribution must follow inverse transforms,
- Mathematical justification for frequency-to-spatial mapping,
- Gradient-based explainability in frequency space,
- Alternative explanation strategies beyond gradients,
- Practical considerations, pitfalls, and validation methods.

2 Key Principle: Attribution Must Follow the Inverse Transform

2.1 Problem Setup

Let:

$$X \in \mathbb{R}^{H \times W}$$

be the spatial image, and

$$X_f = \mathcal{F}(X)$$

be its frequency-domain representation using FFT, DCT, or a similar transform.

The model operates as:

$$X \xrightarrow{\mathcal{F}} X_f \xrightarrow{g(\cdot)} y$$

where g represents the neural network.

2.2 Gradient Propagation and the Adjoint Operator

The gradient of the output with respect to the spatial input is:

$$\frac{\partial y}{\partial X} = \frac{\partial y}{\partial X_f} \cdot \frac{\partial X_f}{\partial X}$$

Since $X_f = \mathcal{F}(X)$, we have:

$$\frac{\partial X_f}{\partial X} = \mathcal{F}$$

Thus:

$$\frac{\partial y}{\partial X} = \frac{\partial y}{\partial X_f} \cdot \mathcal{F}$$

Because gradients live in the dual space, the correct mapping is via the adjoint operator:

$$\boxed{A_s = \mathcal{F}^*(A_f)}$$

where:

$$A_f = \frac{\partial y}{\partial X_f}, \quad A_s = \frac{\partial y}{\partial X}$$

For unitary transforms (FFT with orthonormal normalization):

$$\mathcal{F}^* = \mathcal{F}^{-1}$$

Thus:

$$\boxed{A_s = \mathcal{F}^{-1}(A_f)}$$

2.3 Interpretation

- Gradients in frequency space indicate *which frequencies influence the output*.
- The inverse transform maps this importance back to pixel space.
- This preserves causality and mathematical correctness.

3 Gradient-Based Explainability in Frequency Domain

3.1 Pipeline

1. Forward pass:

$$X \rightarrow X_f = \mathcal{F}(X) \rightarrow y$$

2. Compute gradient:

$$A_f = \left| \frac{\partial y}{\partial X_f} \right|$$

3. Map to spatial domain:

$$A_s = |\mathcal{F}^{-1}(A_f)|$$

4. Normalize and visualize

3.2 Why This Works

- Gradients indicate sensitivity of output to frequency coefficients.
- Inverse FFT maps this sensitivity back to pixel space.
- Preserves causality and respects signal structure.

3.3 Important Properties

- Phase carries spatial localization.
- Magnitude carries energy.
- Discarding phase leads to blurred or misleading explanations.
- FFT normalization must be consistent.

4 Alternative Explainability Methods for Frequency-Based Models

4.1 1. Frequency Perturbation / Occlusion

Idea: Remove or perturb frequency components and observe change in output.

$$\Delta y = y(X_f) - y(\tilde{X}_f)$$

Types:

- Low-pass / high-pass removal
- Band-stop filtering
- Phase randomization
- Directional masking

Advantages:

- Causal
- Model-agnostic
- Interpretable

Disadvantages:

- Computationally expensive
- Limited spatial localization

4.2 2. Energy-Based Attribution

Define energy:

$$E(f) = |X_f(f)|^2$$

Contribution:

$$C(f) = \frac{\partial y}{\partial E(f)} \cdot E(f)$$

Advantages:

- Stable
- Noise-resistant
- Physically interpretable

Applications:

- Texture bias analysis
- Medical imaging
- Signal processing

4.3 3. Spectral TCAV (Concept-Based)

Defines frequency concepts:

- Low frequency
- Mid frequency
- High frequency

Measures:

$$\text{TCAV}(c) = \mathbb{E} \left[\mathbb{1} \left(\frac{\partial y}{\partial v_c} > 0 \right) \right]$$

Benefits:

- Human-interpretable
- Class-level explanation
- Stable across runs

4.4 4. Frequency Response Analysis

Inject sinusoidal signals:

$$X(x, y) = \sin(2\pi f_x x + 2\pi f_y y)$$

Measure output response to obtain:

- Frequency sensitivity
- Directional preference
- Band-pass characteristics

Analogous to a system transfer function.

4.5 5. Wavelet-Based Explainability

- Multi-scale
- Localized in space and frequency
- Avoids FFT globality

Procedure:

$$X \rightarrow \text{Wavelet Decomposition} \rightarrow \text{CNN}$$

Attributions computed per scale and reconstructed spatially.

4.6 6. Information-Theoretic Attribution

Compute mutual information:

$$I(Y; X_f^{(k)})$$

Pros:

- Model-agnostic
- Causal

Cons:

- Computationally expensive
- Difficult to estimate accurately

4.7 7. Hybrid Spatial–Frequency XAI

Best practical approach:

1. Compute frequency attribution
2. Inverse transform to spatial
3. Fuse with spatial Grad-CAM
4. Validate using ablations

5 Comparison of Methods

Method	Causal	Spatial	Stable	Interpretable	Cost
Grad + IFFT	$\triangle$	$\checkmark$	$\triangle$	Medium	Low
Frequency Occlusion	$\checkmark$	$\triangle$	$\checkmark$	High	High
Energy Attribution	$\triangle$	$\times$	$\checkmark$	High	Low
TCAV (Freq)	$\checkmark$	$\times$	$\checkmark$	Very High	Medium
Wavelet XAI	$\triangle$	$\checkmark$	$\checkmark$	High	Medium
Frequency Response	$\checkmark$	$\times$	$\checkmark$	Medium	High

6 Best-Practice Recommendation

For research-quality explainability:

1. Use frequency occlusion for causality
2. Use gradient + inverse FFT for visualization
3. Use wavelet analysis for localization
4. Validate via ablation and perturbation
5. Report frequency sensitivity curves

7 Key Insight

Frequency-domain XAI explains not *where* the model looks, but *what kind of structure it relies on*.

Spatial explanations describe localization. Frequency explanations describe representation bias. Both are necessary for trustworthy interpretability.